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Presence of *Candidatus* Neoehrlichia mikurensis and *Babesia microti* in rodents and two tick species (*Ixodes ricinus* and *Ixodes trianguliceps*) in Slovakia

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ABSTRACT

Rodents are important reservoir hosts of many tick-borne pathogens. Their importance in the circulation of the emerging bacterial agent, *Candidatus* Neoehrlichia mikurensis and the intraerythrocytic protozoan parasite, *Babesia microti* has been recently proposed. The aim of the present study was to identify the presence and genetic diversity of *Candidatus* N. mikurensis and *B. microti* circulating in the natural foci among rodents and two species of ixodid ticks (*Ixodes ricinus* and *Ixodes trianguliceps*). In 2011–2013, rodents were captured at sampling sites in Eastern Slovakia. A total of 997 rodents (324 *Apodemus agrarius*, 350 *Apodemus flavicollis*, 271 *Myodes glareolus*, and 52 other rodent species), 788 feeding ticks from rodents, and 1375 questing ticks were investigated for the presence of pathogens by molecular methods followed by DNA sequencing. *Candidatus* N. mikurensis was detected in 2.4% of questing *I. ricinus* nymphs and 2.6% of questing adult *I. ricinus* ticks, spleens of rodents (1.6%), as well as in feeding larval *I. ricinus* (0.3%) and feeding larval *I. trianguliceps* ticks (3.3%). The 16S rRNA and *glt*A gene sequences of *Candidatus* N. mikurensis obtained in this study confirmed a high degree of genetic identity of this bacterium in Europe. DNA of *B. microti* was found in ear (0.6%) and spleen biopsies of rodents (1.9%), in rodent foetus (3.8%) and feeding larval (5.2%) and nymphal (8.7%) *I. ricinus*, in questing nymphal *I. ricinus* (0.5%) and questing adult *I. ricinus* ticks (0.3%). None of the 112 *I. trianguliceps* ticks were infected. *B. microti* was represented by two different genotypes: 92% of the positive samples belonged to the zoonotic type strain from Jena (Germany). The results of this study underline the importance of rodents in the circulation of both emerging pathogens in natural foci.

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Introduction

In Europe, *Candidatus* N. mikurensis represents a newly emerging tick-borne zoonotic bacterium from the family Anaplasmataceae (Kawahara et al., 2004). It is an obligate intracellular Gram-negative organism, transmitted by *Ixodes* ticks in temperate regions of the Northern hemisphere (Schouls et al., 1999; Vichova et al., 2014; Derdakova et al., 2014). Recently, Burri et al. (2014) confirmed the reservoir competence of rodents for this bacterium in xenodiagnostic experiments. Moreover, it was detected in the blood and endothelial cells of rodent spleens and livers proving systemic infection (Jahfari et al., 2012; Burri et al., 2014). *Candidatus* N. mikurensis could not be cultivated in vitro so far, therefore, the status "*Candidatus*" is still preserved. Since 2007 in Europe, only eight human cases of disease were reported, mainly from immunocompromised patients (Welc-Faleciak et al., 2014).

Babesia spp. represent widespread protozoan hemoparasites from the family Babesiidae. *Ixodes ricinus* plays an important role in the transmission of several protozoan species of the genus *Babesia*. *Babesia divergens*, *Babesia venatorum*, and *Babesia microti* occur in Europe and are proven aetiological agents of human

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babesiosis (Meer-Scherrer et al., 2004; Hildebrandt et al., 2007; Gray et al., 2010; Holler et al., 2013). Usually, the course of babesiosis caused by *B. microti* is asymptomatic or mild, but it may occur as a severe infection in splenectomized patients. Circulation of *B. microti* in European natural foci is primarily associated with small rodents. Microtine rodents and probably shrews are the most important reservoir hosts (Bown et al., 2008; Silaghi et al., 2012; Nefedova et al., 2013). Phylogenetic analyses confirmed that *B. microti* is a complex species, consisting of genetically diverse isolates that form different clades based on the analysed genes (Goethert and Telford, 2003; Nakajima et al., 2009). Within these clades, rodent isolates are subdivided into the zoonotic"Jena" type (GenBank accession number EF413181) (Hildebrandt et al., 2007) and the non-zoonotic "Munich" type (GenBank accession number AB071177) strains (reviewed in Gray et al., 2010). Different *B. microti* strains may circulate in a given natural focus in different rodent species at the same time. There are assumptions that in the circulation of human pathogenic ecotypes, the yellow-necked mouse (*Apodemus flavicollis*), which is very frequent in natural foci of Slovakia, might be predominantly involved (Stanko, 1989, 1995).

The main aims of our study were to monitor the infection prevalence and genetic diversity of *Candidatus* N. mikurensis and *B. microti* in order to elucidate their ecological associations with rodents, questing *I. ricinus*, and feeding ticks (*I. ricinus* and *Ixodes trianguliceps*) collected from rodents at four selected sites of Slovakia.

Materials and methods

Study area

The study was conducted at four sampling sites in Eastern Slovakia (Čermel', Hýľov, Botanical garden Košice, and Rozhanovce). Two sites, Čermel' (208–600 m above sea level (asl); 48°45′46.67″N; 21°8′8.17″E) and Hýľov (500–750 m asl; 48°44′22.80″N; 21°4′18.90″E) were characterized by the presence of two ixodid tick species, *I. ricinus* and *I. trianguliceps*; whereas the two other sites, Botanical garden Košice (208 m asl; 48°44′6.84″N; 21°14′16.14″E) and Rozhanovce (215 m asl; 48°45′00″N; 21°21′00″E), were characterized by the exclusive presence of *I. ricinus* and absence of *I. trianguliceps* ticks. For the landscape in Čermel', the presence of sylvatic deciduous forest is typical, and Hýľov is characterized by sylvatic mixed forest. The locality Botanical garden is a typical suburban mixed forest. Rozhanovce is a game reserve in a deciduous forest with the presence of mouflon (*Ovis musimon*) and fallow deer (*Dama dama*) herds (Blaňarová et al., 2014).

Sample collection

Questing *I. ricinus* nymphs and adults were collected twice a year (June and September) during two consecutive years 2011–2012 at four selected sites in Eastern Slovakia. Ticks were collected at each study site by flagging with a 1-m^2 white corduroy cloth for one hour to cover various types of forest/shrubland and edge vegetation. Immediately after the collection, ticks were identified to species and sex according to morphological key (Filippova, 1977; Estrada-Peña et al., 2004) and preserved in 70% ethanol until the DNA was extracted.

Trapping of rodents and removal of rodent-fed ticks were performed during 2011–2013 at four sampling sites in Eastern Slovakia. Rodents were trapped alive using Swedish bridge metal traps following the protocol of Stanko (1995). Rodent trappings were carried out using 100–150 live-traps/per site for two consecutive nights. Small mammals, rodents and insectivores, were euthanized in a humane manner in accordance with the laws of the Slovak Republic under the licenses of the Ministry of Environment of the Slovak Republic No. 297/108/06-3.1 and No. 6743/2008-2.1. Rodent-fed ticks were removed with sterile forceps, counted, and identified to life stage and species using previously published taxonomic keys (Filippova, 1977; Estrada-Peña et al., 2004) and preserved in 70% ethanol until DNA extraction. A maximum of 20 feeding and visibly engorged *I. ricinus* ticks per captured rodent and all feeding *I. trianguliceps* ticks were analysed. Moreover, spleen and ear biopsies were obtained from each rodent and rodent foetuses from gravid females during necroscopy were stored at −20 °C prior to use. Age of rodent foetuses was estimated using the study of Pelikán (1966), assuming the pregnancy of *A. flavicollis* and *Apodemus agrarius* takes approximately 23–24 days.

DNA extraction

Questing ticks and ticks removed from rodents were further analysed by DNA based methods. Genomic DNA was isolated from individual ticks by alcaline-hydrolysis according to Guy and Stanek (1991). DNA from rodent tissues was extracted using a commercial DNA extraction kit (NucleoSpin Blood kit, NucleoSpin Tissue kit, Machery Nagel, Germany) according to the manufacturer's protocol. Lysates were stored at −20 °C.

Molecular detection of pathogens

Polymerase chain reaction (PCR) amplification of the tick mitochondrial *cytochrome b* gene and 12S rRNA gene of vertebrates was performed for each sample as a quality control for extracted tick and tissue DNA, respectively (Black and Roehrdanz, 1998; Derdakova et al., 2003; Humair et al., 2007).

Only positive samples were further screened for the presence of *Candidatus* N. mikurensis by real-time PCR using the primers NMGroELF, NMGroELR and the NMikGroEL-P2a TaqMan-Probe labelled with Texas Red, which targeted a 99-bp long fragment of the *groEL* gene (Jahfari et al., 2012). This assay was run on a CFX96 Real-Time PCR System (Bio-Rad, Hercules, CA, USA). To further characterize *Candidatus* N. mikurensis-positive samples, two molecular loci, 16S rRNA and *gltA* were amplified and sequenced. Single PCRs were performed to amplify a 560-bp fragment of the 16S rRNA gene using the primers IS58-594r and IS58-62f as previously described (Kawahara et al., 2004) and a 233-bp fragment of the *gltA* gene with the primers NMikfogltA, NMikregltA (Jahfari et al., 2012).

For the molecular detection of *Babesia* spp., PCR amplification of a 450-bp fragment of the small 18S subunit of ribosomal RNA with primers BJ1 and BN2 was performed (Casati et al., 2006).

The PCR reactions were performed in a total volume of 25 μl of reaction mixture with 12.5 μl 1× Mix, 2.6 μl 2.5 mM $MgCl_2$, 2.3 μl 920 nM of each primers, and 0.3 μl 0.12 nM of probe using 5 μl as DNA template. Kit Bioron SuperHot Master Mix (Bioron, Germany) was used for real-time PCR and kit HotStarTaq Plus DNA Polymerase (Qiagen) for PCR analyses. In each PCR, previously sequenced DNA samples of *Candidatus* N. mikurensis and/or *B. microti* were used as positive controls. Nuclease-free water was added in PCR mix instead of tested DNA as negative control. The PCR products were visualized by electrophoresis on 2% agarose gels stained with GoldView Nucleic Acid Stain (Beijing SBS Genetech, Beijing, China). All positive PCR products were purified by QIAquick PCR purification kit (Qiagen) and bidirectionally Sanger-sequenced with the same primers as for the PCR amplifications.

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Sequence analysis

Nucleotide sequences were manually edited in MEGA 5 (Tamura et al., 2011) and compared with GenBank entries by BLAST (Altschul et al., 1997).

The nucleotide sequence data reported in this paper are available in GenBank database under the following accession numbers: *Candidatus* N. mikurensis 16S rRNA (KJ649314–KJ649316, KJ649323), *Candidatus* N. mikurensis *glt*A (KJ649317–KJ322) and *B. microti* 18S rRNA (KJ649287–KJ649312).

Statistical analysis

A 95% confidence intervals (CI) and differences in *Candidatus* N. mikurensis and *B. microti* infection prevalences among sites were tested by Fisher's exact test and calculated by the Quantitative Parasitology on the Web (Reiczigel et al., 2005) [http://www.univet.hu/qpweb/qp10/index.php].

Results

Candidatus *N. mikurensis in rodents*

Candidatus N. mikurensis was not detected in any of 790 tested ear biopsies of rodents, but it was confirmed in ten out of 621 tested spleens (1.6%). Five positive spleen biopsies were obtained from rodents trapped in Čermel' (5/193, 2.6%; CI: 0.84–5.95), and five spleens were found at the sampling site Hýl'ov (5/81, 6.2%; CI: 2.03–13.83) (Table 1). *Candidatus* N. mikurensis was detected in the spleens of *A. flavicollis* (6/350; 1.7%; CI: 0.63–3.7), in *Myodes glareolus* (3/271; 1.1%; CI: 0.22–3.21), and *A. agrarius* (1/324; 0.3%; CI: 0.00–1.71) (Table 2).

These three rodent species were also the most commonly trapped rodents during our study. Positive samples from rodents came only from two control sites, Čermel' and Hýl'ov, where *I. trianguliceps* and *I. ricinus* ticks co-occur. We found no significant differences between the rates of infection in spleen samples among the four sites (*F*-value 0.167, $p > 0.05$).

Moreover, *Candidatus* N. mikurensis was detected in a foetus of a negative pregnant *A. flavicollis* female (overall prevalence: 1.9%; 1/53) (Table 1).

Candidatus *N. mikurensis in rodent-fed ticks*

A total of 788 feeding ticks (112 *I. trianguliceps* and 676 *I. ricinus*) were tested. Three larval *I. trianguliceps* (3/112; 2.7%) and two larval *I. ricinus* (2/676; 0.3%) ticks were positive for the presence of *Candidatus* N. mikurensis with an overall prevalence of 0.6% (Table 1). Positive/infected feeding *I. trianguliceps* larvae (3/54; 5.6%, CI: 1.16–15.39) were collected from *M. glareolus* and *A. agrarius* from the site Čermel', and positive *I. ricinus* larvae (2/348; 0.6%, CI: 0.06–2.07) from *A. agrarius* from Botanical garden in Košice (Tables 2 and 3). We found no significant difference in *Candidatus* N. mikurensis prevalence in rodent-fed ticks among the four study sites (*F*-value 0.051, $p > 0.05$). In two *I. ricinus* larvae a co-infection of *Candidatus* N. mikurensis and *B. microti* was detected.

Candidatus *N. mikurensis in questing* I. ricinus *ticks*

In total, 1375 *I. ricinus* ticks from vegetation were tested for the presence of *Candidatus* N. mikurensis by real-time PCR targeting the *groEL* gene resulting in an overall *Candidatus* N. mikurensis infection prevalence of 2.5% (34/1375; CI: 1.71–3.44). *Candidatus* N. mikurensis was detected in 2.4% of questing nymphal *I. ricinus* and 2.6% of questing adult *I. ricinus* ticks. We found no significant

Table 1
Number of positive and tested questing, feeding ticks and rodent biopsies (ear, spleen, foetus) positive for *Candidatus* N. mikurensis by PCR, and prevalences of infection (%).

Site	Questing ticks		Feeding ticks				Biopsies		
	No. of positive nymphal IR/n (prevalence in %)	No. of positive adult IR/n (prevalence in %)	No. of positive larval IR ticks/no. of larval IR ticks; prevalence (%)	No. of positive nymphal IR ticks/no. of nymphal IR ticks; prevalence (%)	No. of positive larval IT ticks/no. of larval IT ticks; prevalence (%)	No. of positive nymphal IT ticks/no. of nymphal IT ticks; prevalence (%)	No. of positive female IT ticks/no. of female IT ticks; prevalence (%)	No. of positive ear/no. of ear; prevalence (%)	No. of positive spleen/no. of spleen; prevalence (%)	No. of positive rodent foetus/no. of foetus; prevalence (%)
Čermel'	1/123 (0.8)	0/96	0/25	0/1	3/54 (5.6)	0/6	0/3	0/173	5/193 (2.6)	1/34 (2.9)
Hýl'ov	5/188 (2.7)	6/78 (7.7)	0/104	0/4	0/33	0/6	0/3	0/85	5/81 (6.2)	0
B. garden Košice	2/137 (1.5)	1/39 (2.6)	2/348 (0.6)	0/27	0	0	0	0/45	0/33	0/8
Košice Rozhanovce	17/583 (2.9)	2/131 (1.5)	0/153	0/14	0/4	0/3	0	0/487	0/314	
Total	25/1031 (2.4)	9/344 (2.6)	2/630 (0.3)	0/46	3/91 (3.3)	0/15	0/6	0/790	10/621 (1.6)	1/53 (1.9)

IR—*I. ricinus*, IT—*I. trianguliceps*.

Table 2
Number of rodents positive for *Candidatus* N. mikurensis by PCR and number of trapped rodents used for molecular analysis at selected sites in 2011–2013.

Species of rodents	No. of positive rodents/no. of total trapped rodents/prevalence (%)				
	Čermel'	Hýľov	B. garden Košice	Rozhanovce	Total
Apodemus agrarius	0/31	1/8 (12.5)	0/54	0/231	1/324 (0.3)
Apodemus flavicollis	2/88 (2.3)	4/55 (7.3)	0/26	0/181	6/350 (1.7)
Myodes glareolus	3/127 (2.4)	0/39	0	0/105	3/271 (1.1)
Crocidura suaveolens	0	0	0	0/2	0/2
Micromys minutus	0	0	0/1	0	0/1
Microtus arvalis	0/1	0	0	0/10	0/11
Microtus subterraneus	0/1	0/3	0/1	0/2	0/7
Neomys fodiens	0/2	0	0	0	0/2
Rattus norvegicus	0	0	0	0/10	0/10
Sorex araneus	0/8	0/3	0/1	0/4	0/16
Sorex minutus	0/1	0	0	0/2	0/3
Total	5/259 (1,9)	5/108 (4.6)	0/83	0/547	10/997 (1.0)

differences between the infection rates from different sites (*F*-value 0.052, $p > 0.05$) (Table 1).

Sequence analysis: 16S rRNA and gltA gene sequences of Candidatus *N. mikurensis*

In samples which tested positive for *Candidatus* N. mikurensis *groEL* gene, we further amplified and sequenced two other genetic loci, 16S rRNA and *gltA*. Four *Candidatus* N. mikurensis 16S rRNA (KJ649314–KJ649316, KJ649323) and six *Candidatus* N. mikurensis *glt*A (KJ649317–KJ322) sequences were deposited in GenBank database (Table 3).

The obtained *Candidatus* N. mikurensis 16S rRNA sequences from rodent spleen biopsies, from feeding ticks removed from rodents, and from *I. ricinus* ticks from vegetation showed 100% identity with the various *Candidatus* N. mikurensis isolates deposited in GenBank, e.g. an isolate from an *I. persulcatus* tick from Russia (FJ966360), nucleotide sequences from rodents from China (JQ359052) and Japan (AB196305), that were also identical with a human isolate from China (JQ359045).

The nucleotide sequences of *Candidatus* N. mikurensis *gltA* gene from rodent spleen biopsies, from ticks feeding on rodents and from *I. ricinus* ticks from vegetation showed 100% homology with the *Candidatus* N. mikurensis isolate taken from an *I. ricinus* tick from Italy (AY098729).

B. microti in rodents

B. microti was detected in five out of 784 tested ear biopsies (0.6%) and in twelve out of 625 tested spleens (1.9%) of rodents (Table 4). One positive spleen (1/190, 0.5%; CI: 0.01–2.90) and one positive ear (1/161, 0.6%; CI: 0.01–3.42) came from rodents trapped in Čermel', one positive ear sample was obtained in Hýľov (1/89, 1.1%; CI: 0.02–6.11), two positive spleens (2/44, 5%; CI: 0.55–15.48) and eight ear samples (8/37, 22%; CI: 9.82–38.22) originated from Botanical garden, and three spleens (3/320, 0.9%; CI: 0.19–2.72) and one ear sample (1/490, 0.2%; CI: 0.00–1.14) came from Rozhanovce (Table 4). Overall, the pathogen was detected in twelve biopsies from *A. agrarius* (12/324; 3.7%; CI: 1.92–6.39), in four biopsies from *A. flavicollis* (4/350; 1.1%; CI: 0.31–2.91), and in three biopsies from *M. glareolus* (3/271; 1.1%; CI: 0.22–3.21) (Table 5). These three rodent species were also the most commonly trapped rodents during the study. We found significant differences between the infection rates in ear biopsies (*F*-value 0.016, $p < 0.05$) and spleens (*F*-value $p < 0.001$) from different sites.

B. microti was also detected in two rodent foetuses from the site Botanical garden with an overall prevalence of 4% (2/53). One positive foetus (6.0 mm, 12 days old) was detected in uninfected *A. flavicollis* female. Second positive foetus (15.0 mm, 17 days old) was detected in one out of four infected *A. agrarius* female (Table 4).

Table 3
Accession numbers (GenBank database) of *Candidatus* N. mikurensis sequences for the 16S rRNA and *gltA* genes generated in this study.

Host, vector	Sequences of 16S rRNA gene	Sequences of *gltA* gene
Rodent foetus from *A. flavicollis*	KJ649314	[a]
Feeding *I. ricinus* larva, *A. agrarius*	[a]	KJ649321
Feeding *I. ricinus* larva, *A. agrarius*	[a]	KJ649322
Feeding *I. trianguliceps* larva, *A. agrarius*	KJ649315	[a]
Questing *I. ricinus* male	KJ649316	KJ649319
Questing *I. ricinus* nymps	[a]	KJ649320
Spleen of *A. agrarius*	[a]	KJ649317
Spleen of *A. flavicollis*	KJ649323	KJ649318

[a] Were not sequenced.

B. microti in rodent-feeding ticks

A total of 788 feeding ticks were tested, and 37 *I. ricinus* ticks (5.5%) tested positive for *B. microti* (Table 4). *B. microti*-positive *I. ricinus* ticks (32 larvae and 3 nymphs) were collected from *A. agrarius* and one larva and nymph were removed from *A. flavicollis* from the site Botanical garden in Košice (Tables 5 and 6).

B. microti in questing I. ricinus *ticks*

Altogether 1368 questing *I. ricinus* ticks were tested for the presence of *B. microti* by PCR targeting the 18S rRNA gene resulting in an overall *B. microti* infection prevalence of 0.4% (6/1368; CI: 0.16–0.96). DNA of *B. microti* was found in questing nymphal *I. ricinus* (0.5%) and questing adult *I. ricinus* ticks (0.3%). The statistical analysis revealed significant differences among the tick infection rates from different sites (*F*-value 0.036, $p < 0.05$).

Sequence analysis: 18S rRNA gene sequences of Babesia spp.

All *Babesia*-positive PCR products were further sequenced and identified as *B. microti.* In total, 26 obtained 18S rRNA sequences were deposited in GenBank database (KJ649287–KJ649312). A BLAST analysis confirmed the genotype of the "zoonotic strain" (GenBank accession number EF413181) in 24 *B. microti*-positive samples (92%; KJ649287–KJ649290, KJ649292–KJ649301, KJ649303–KJ649312) and the "non-zoonotic" strain (GenBank accession number AB071177) in two *B. microti* isolates (8%; KJ649291, KJ649302) (Table 6). The zoonotic strain nucleotide sequences were 100% identical with human isolates from Poland

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Table 4
Number of positive and tested questing, feeding ticks and rodent biopsies (ear, spleen, foetus) positive for *Babesia microti* by PCR, and prevalences of infection (%).

Site	Questing ticks		Feeding ticks				Biopsies		
	No. of positive nymphal IR/n (prevalence in %)	No. of positive adult IR/n (prevalence in %)	No. of positive larval IR ticks/no. of larval IR ticks; prevalence (%)	No. of positive nymphal IR ticks/no. of nymphal IR ticks; prevalence (%)	No. of positive larval IT ticks/no. of larval IT ticks; prevalence (%)	No. of positive nymphal IT ticks/no. of nymphal IT ticks; prevalence (%)	No. of positive female IT ticks/no. female IT ticks; prevalence (%)	No. of positive ear/no. of ear; prevalence (%)	No. of positive spleen/no. of spleen; prevalence (%)	No. of positive rodent foetus/no. of foetus; prevalence (%)
Čermel'	0/123	0/89	0/25	0/1	0/54	0/6	0/3	1/161 (0.6)	1/190 (0.5)	0/34 (2.9)
Hýľov	3/188 (1.6)	0/78	0/104	0/4	0/33	0/6	0/3	1/89 (1.1)	0/78	0
B. garden Košice	1/137 (0.7)	1/39 (2.6)	33/348 (9.5)	4/27 (14.8)	0	0	0	2/44 (4.6)	8/37 (21.6)	2/8 (25.0)
Rozhanovce	1/583 (0.2)	0/131	0/153	0/14	0/4	0/3	0	1/490 (0.2)	3/320 (0.9)	0/11
Total	5/1031 (0.5)	1/337 (0.3)	33/630 (5.2)	4/46 (8.7)	0/91	0/15	0/6	5/784 (0.6)	12/625 (1.9)	2/53 (3.8)

IR—*I. ricinus*, IT—*I. trianguliceps*.

(KC470047), Germany (EF413181), or with a sequence from *I. ricinus* collected in Germany (JX679169). The "non-zoonotic"strain showed 100% identity with a rodent isolate from Japan (AB071177) or 99% identity with an *I. ricinus* tick sequence from Poland (AY789075).

B. microti isolates from Slovakia clustered with genotypes from North America and Europe (Fig. 1).

Discussion

In the present study, we investigated the prevalence of infection and diversity of *Candidatus* N. mikurensis and *B. microti* in rodents, in questing and in feeding ticks (*I. ricinus* and *I. trianguliceps*) collected from rodents. We conducted a comparative study in two sites known for the occurrence of *I. ricinus* only (Botanical garden Košice, Rozhanovce) and two sites at a higher altitude (600–750 m asl) (Čermel' and Hýľov) where both tick species (*I. ricinus* and *I. trianguliceps*) occur (Petko et al., 1991). In this study, we did not find significant differences among the infection prevalence of *Candidatus* N. mikurensis in spleen samples and questing *I. ricinus* ticks from different sites. In contrast, we found significant differences among the prevalence of *B. microti* in ear and spleen biopsies and questing *I. ricinus* ticks from different sites. The highest prevalence of *B. microti* in rodent biopsies was observed in urban park Botanical garden that might indicate the presence of suitable reservoir hosts in urban areas.

The overall prevalence of *Candidatus* N. mikurensis infection in rodent spleen biopsies (*M. glareolus* and *Apodemus* spp.) in our study was 1.6%. Other authors from Europe reported the rate of infection in different rodent organs (kidney, liver, skin, and brain) varies from 1.0% up to 52.7% (Jahfari et al., 2012; Silaghi et al., 2012; Burri et al., 2014). *Candidatus* N. mikurensis was detected at selected sites where *I. trianguliceps* ticks were present. None of the 790 rodent ear biopsies tested positive. In eco-epidemiological studies of Silaghi et al. (2012), different animal organs were used as the source of genomic DNA extraction and further detection of *Candidatus* N. mikurensis DNA. After experimental infection of rats, no infection was observed after 15 days in blood, spleen or liver, a weak infection was detected in the spleen after 30 days, and in all three organs 60 days after initial infection. This is in accordance with our results that that spleen and kidneys are the most appropriate organs for detection of *Candidatus* N. mikurensis in rodents. As confirmed in present study, skin biopsies are not suitable for molecular identification of *Candidatus* N. mikurensis.

A recent study conducted by Burri et al. (2014) using xenodiagnosis, found that *M. glareolus* and *A. flavicollis* are reservoirs of *Candidatus* N. mikurensis. Similarly as in our study, the bacterium was most commonly found in the spleens of *A. flavicollis* (60%), followed by *M. glareolus* (30%) and *A. agrarius* (10%). Similar results were obtained by Vichova et al. (2014). Insectivores in Germany were found to be negative for the presence of *Candidatus* N. mikurensis (Silaghi et al., 2012), however urban hedgehogs (*E. roumanicus*) in Hungary were found to be carriers of *Candidatus* N. mikurensis. It still remains to be determined whether the hedgehog, a non-rodent species, is a reservoir of *Candidatus* N. mikurensis. Moreover, we detected *Candidatus* N. mikurensis in a foetus of *A. flavicollis*. The pregnant female was in the third week, i.e. in the last week of gravidity. Similar results were reported also by Obiegala et al. (2014).

Candidatus N. mikurensis was detected only in two larvae of *I. ricinus* fed on rodents (0.3%), not in unfed *I. ricinus* larvae. In Europe, *Candidatus* N. mikurensis was detected in partially fed larvae, nymphs and adult *I. ricinus* ticks in Germany and Switzerland with the prevalence ranging from 2.6% to 6.4%, respectively (Silaghi et al., 2012; Jahfari et al., 2012; Burri et al., 2014). In our study,

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Table 5
Number of rodents positive for *B. microti* by PCR and number of trapped rodents used for molecular analysis at selected sites in 2011–2013.

Species of rodents	No. of positive rodents/no. of total trapped rodents/prevalence (%)				
	Čermel'	Hýľov	B. garden Košice	Rozhanovce	Total
Apodemus agrarius	0/31	1/8 (12.5)	11/54 (20.4)	0/231	12/324 (3.7)
Apodemus flavicollis	0/88	0/55	1/26 (3.8)	3/181 (1.7)	4/350 (1.1)
Myodes glareolus	2/127 (1.6)	0/39	0	1/105 (1.0)	3/271 (1.1)
Crocidura suaveolens	0	0	0	0/2	0/2
Micromys minutus	0	0	0/1	0	0/1
Microtus arvalis	0/1	0	0	0/10	0/11
Microtus subterraneus	0/1	0/3	0/1	0/2	0/7
Neomys fodiens	0/2	0	0	0	0/2
Rattus norvegicus	0	0	0	0/10	0/10
Sorex araneus	0/8	0/3	0/1	0/4	0/16
Sorex minutus	0/1	0	0	0/2	0/3
Total	2/259 (0.8)	1/108 (0.9)	12/83 (14.5)	4/547 (0.7)	19/997 (1.9)

co-feeding *I. ricinus* and *I. trianguliceps* ticks were observed on 17 rodents. Sixty-four rodents were infested only with *I. trianguliceps* ticks, however this does not exclude the possibility of previous infestations with *I. ricinus*. Interestingly, we detected the presence of *Candidatus* N. mikurensis DNA in three out of 112 feeding larval *I. trianguliceps* (prevalence of 2.7%). This is the first detection of *Candidatus* N. mikurensis in partially feeding larval *I. trianguliceps* in Europe.

The overall prevalence of *Candidatus* N. mikurensis in questing *I. ricinus* nymphs and adults recorded in our study was 2.5% which is within the range 1–26.6% reported by other authors across Europe (Spitalska et al., 2008; Rar and Golovljova, 2011; Jahfari et al., 2012; Silaghi et al., 2012). The presence of *Candidatus* N. mikurensis was detected at all sampling sites involved in this study.

Genetic variability of *Candidatus* N. mikurensis isolates from questing ticks, from feeding ticks and rodents was studied by DNA amplification and sequencing of two variable gene loci (16S rRNA, *gltA*). In this study, we did not observe any intraspecific variability. Our nucleotide sequences were identical with *Candidatus* N. mikurensis isolates from questing ticks, rodents, and from human patients from Europe and Asia (Brouqui et al., 2003; Rar et al., 2010; Jahfari et al., 2012; Silaghi et al., 2012).

B. microti was detected in *A. agrarius* (3.7%), *A. flavicollis* (1.1%) and *M. glareolus* (1.1%) with an overall prevalence of 1.9%. Our results are similar to the results from other European countries with prevalences varying from 1% up to 45.2% in rodents (Duh et al., 2003; Sinski et al., 2006; Hartelt et al., 2008; Bown et al., 2008; Nefedova et al., 2013). *B. microti* was detected in ear biopsies of rodents at all sampling sites (0.6%) and in rodent spleens at three sites except Hýľov (1.9%). Silaghi et al. (2012) reported the presence of *B. microti* in one case, only in the liver of *A. agrarius* (1/90).

In the present study, we identified *B. microti* in two foetuses of *A. agrarius* and *A. flavicollis* females, and in the spleen sample of *A. agrarius* female. These females were in the second and in the third week of gravidity, respectively.

B. microti was also found in *I. ricinus* larvae (5.2%) and nymphs (8.7%) fed on rodents with an overall prevalence of 4.7%. Surprisingly, in our study none of 112 tested feeding *I. trianguliceps* ticks was *B. microti*-positive, while Nefedova et al. (2013) observed a high prevalence of this pathogen in this tick species. *B. microti* reported in *I. ricinus* ticks from Europe was associated with human disease (Hildebrandt et al., 2007), whereas the strain circulating in British rodents has been previously demonstrated to be transmitted only by *I. trianguliceps* thus, this is also of questionable zoonotic or veterinary importance. Similarly, in two rodent-fed *I. ricinus* larvae we observed a coinfection of *B. microti* with *Candidatus* N. mikurensis.

In Europe, infection rates of *B. microti* in questing ticks are rather low, ranging from 0.9% up to 5.4% (Rudolf et al., 2005; Sreter et al., 2005; Wojcik-Fatla et al., 2006; Reye et al., 2010, 2013). In our study, babesiae were present in questing *I. ricinus* nymphs and adult ticks

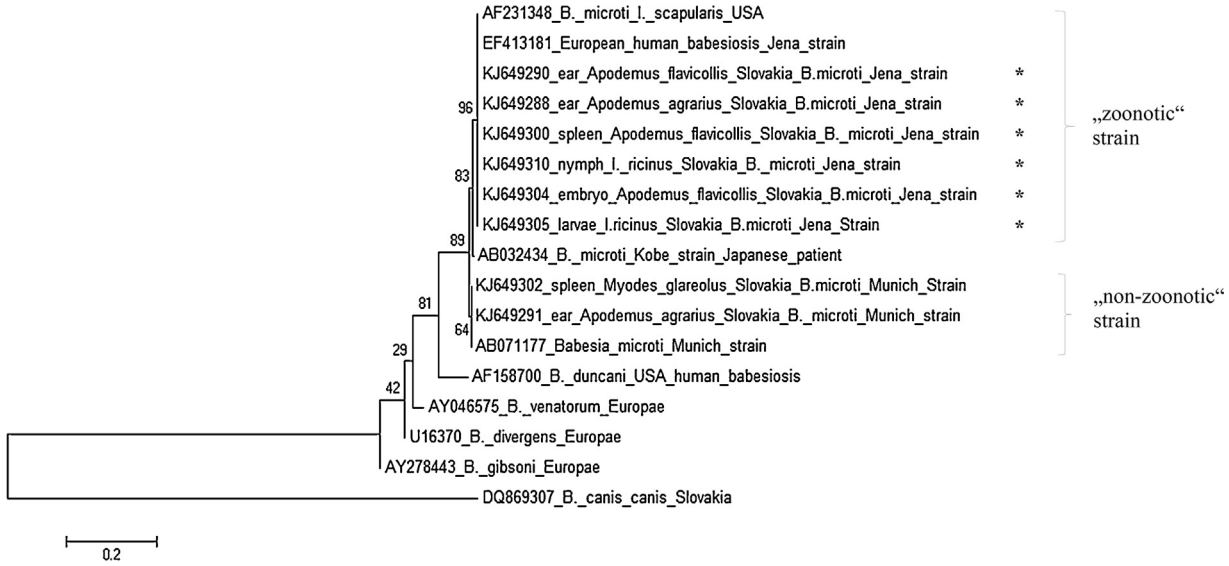


Fig. 1. The distance tree inferred by Neighbor-Joining analysis using the Kimura 2-parameter model of 445 bp long 18S rRNA gene sequences of *B. microti* isolates, *Babesia duncani*, *Babesia venatorum*, *Babesia divergens*, *Babesia gibsoni* and *Babesia canis canis* (outgroup). The numbers at the nodes are bootstrap values expressed as percentages of 1000 bootstrap replicates; the bar (0.2) represents the number of mutations per site. Samples of this study are indicated in mark *.

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Table 6
Accession numbers (GenBank database) of *B. microti* and *Hepatozoon* spp. sequences of the 18S rRNA gene generated in this study.

Host, vector	Sequences of 18S rRNA gene (*Babesia* species)	Sequences of 18S rRNA (*Hepatozoon* species)
Questing *I. ricinus* nymph	KJ649287	[a]
Ear biopsy from *A. agrarius*	KJ649288	[a]
Ear biopsy from *A. agrarius*	KJ649289	[a]
Ear biopsy from *A. flavicollis*	KJ649290	[a]
Ear biopsy from *A. agrarius*	KJ649291	[a]
Spleen from *A. agrarius*	KJ649292	[a]
Spleen from *A. agrarius*	KJ649293	[a]
Spleen from *A. agrarius*	KJ649294	[a]
Spleen from *A. agrarius*	KJ649295	[a]
Spleen from *M. glareolus*	KJ649296	[a]
Spleen from *A. flavicollis*	KJ649297	[a]
Spleen from *A. agrarius*	KJ649298	[a]
Spleen from *A. flavicollis*	KJ649299	[a]
Spleen from *A. flavicollis*	KJ649300	[a]
Spleen from *A. agrarius*	KJ649301	[a]
Spleen from *M. glareolus*	KJ649302	[a]
Ear biopsy from *A. flavicollis*	KJ649303	[a]
Ear biopsy from *A. agrarius*	KJ649304	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649305	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649306	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649307	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649308	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649309	[a]
Feeding *I. ricinus* nymph, *A. agrarius*	KJ649310	[a]
Feeding *I. ricinus* nymph, *A. agrarius*	KJ649311	[a]
Feeding *I. ricinus* larva, *A. agrarius*	KJ649312	[a]

[a] Were not sequenced.

with a prevalence of 0.4%. The presence of zoonotic strains was confirmed in 92% of the positive samples.

Conclusion

The results of this study confirmed that in natural foci of Slovakia, rodents are infected with *Candidatus* N. mikurensis and *B. microti* and might act as reservoir hosts of both pathogens by infecting the feeding ticks. Both, questing nymphal and adult *I. ricinus* and feeding larval and nymphal *I. ricinus* ticks as well as feeding larval *I. trianguliceps* ticks were found to be infected. To our knowledge, the presence of *Candidatus* N. mikurensis in the rodent-feeding larval *I. trianguliceps* was demonstrated for the first time in Europe. Molecular evidence of *B. microti* in rodent foetuses suggests the possibility of transplacental transmission as an alternative route for the spread of infection among rodents.

Conflict of interest statement

The authors declare no conflict of interest.

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